

Continuous Wave Second Harmonics Generation

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THEORY

Coupled wave equations. Considering diffraction, walk off and nonlinearity, the coupled wave equations are:

$$\begin{aligned}\frac{\partial A_{\text{SH}}}{\partial z} &= \frac{i}{2k_{\text{SH}}} \nabla_{\perp}^2 A_{\text{SH}} + \tan \rho \frac{\partial A_{\text{SH}}}{\partial y} + \frac{i\omega_{\text{SH}}}{2n_{\text{SH}}c} d_{\text{eff}} A_{\text{F}}^2 \exp(-i\Delta k z) \\ \frac{\partial A_{\text{F}}}{\partial z} &= \frac{i}{2k_{\text{F}}} \nabla_{\perp}^2 A_{\text{F}} + \frac{i\omega_{\text{F}}}{n_{\text{F}}c} d_{\text{eff}} A_{\text{SH}} A_{\text{F}}^* \exp(i\Delta k z) \\ \Delta k &= k_{\text{SH}} - 2k_{\text{F}}\end{aligned}\quad (1)$$

For LBO in XY plane, $d_{\text{eff}} = d_{32} \cos \varphi$, where $d_{32} = -0.98 \text{ pm/V}$.

The input gaussian beam is:

$$A(x, y, z) = A_0 \frac{\omega_0}{\omega(z)} \exp\left(-\frac{x^2 + y^2}{\omega^2(z)}\right) \exp\left(ik\left(z + \frac{x^2 + y^2}{2R(z)}\right) - i \arctan\left(\frac{z}{z_R}\right)\right) \quad (2)$$

where ω_0 and $\omega(z)$ is beam radius at $1/e^2$ intensity and z is distance from the beam waist. If the average power is P and the refractive index of the medium is n , other parameters are:

$$\begin{aligned}k &= \frac{2\pi}{\lambda} \\ \omega(z) &= \omega_0 \sqrt{1 + \left(\frac{z}{z_R}\right)^2} \\ R(z) &= z + \frac{z_R^2}{z} \\ z_R &= \frac{\pi\omega_0^2}{\lambda} \\ A_0 &= \sqrt{\frac{4P}{\pi\omega_0^2 n c \epsilon_0}}\end{aligned}\quad (3)$$

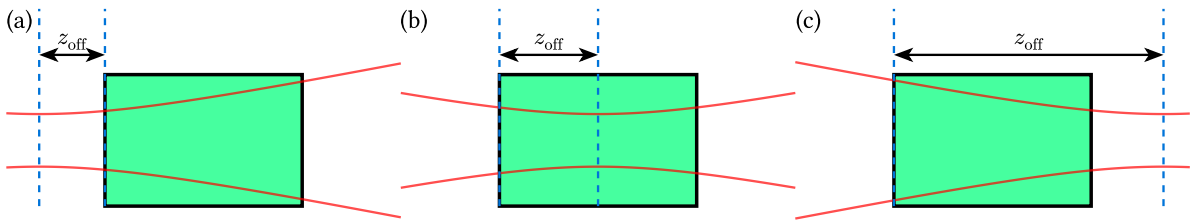


Figure 1: Gaussian beam is focused to the nonlinear crystal

The definition of $D4\sigma$ radius is:

$$\omega_x = 2\sqrt{\frac{\int x^2 I(x, y) dx dy}{\int I(x, y) dx dy}} \quad (4)$$

Numerical algorithm. Equation 1 is solved by using RK4IP method[1], but without adaptive step size control. Equation 1 can be organized as follows:

$$\begin{aligned}\frac{\partial A_S}{\partial z} &= \mathcal{D}_S A_S + \mathcal{N}_S(A_S, A_F) \\ \frac{\partial A_F}{\partial z} &= \mathcal{D}_F A_S + \mathcal{N}_F(A_S, A_F)\end{aligned}\tag{5}$$

where $\mathcal{D}$ represents linear effects and $\mathcal{N}$ represents nonlinear coupling. Then transform A_S and A_F to the interaction picture:

$$\begin{aligned}A_S^{\text{ip}} &= \exp(-(z - z')\mathcal{D}_S)A_S \\ A_F^{\text{ip}} &= \exp(-(z - z')\mathcal{D}_F)A_F\end{aligned}\tag{6}$$

The coupled equations in the interaction picture are:

$$\begin{aligned}\frac{\partial A_S^{\text{ip}}}{\partial z} &= \exp(-(z - z')\mathcal{D}_S)\mathcal{N}_S\left(\exp((z - z')\mathcal{D}_S)A_S^{\text{ip}}, \exp((z - z')\mathcal{D}_F)A_F^{\text{ip}}\right) \\ \frac{\partial A_F^{\text{ip}}}{\partial z} &= \exp(-(z - z')\mathcal{D}_F)\mathcal{N}_F\left(\exp((z - z')\mathcal{D}_S)A_S^{\text{ip}}, \exp((z - z')\mathcal{D}_F)A_F^{\text{ip}}\right)\end{aligned}\tag{7}$$

By choosing $z' = z + \frac{h}{2}$, the Equation 7 can be solved by RK4 method:

$$\begin{aligned}\alpha_{1S} &= \exp\left(\frac{h}{2}\mathcal{D}_S\right)\mathcal{N}_S(A_S, A_F) \\ \alpha_{1F} &= \exp\left(\frac{h}{2}\mathcal{D}_F\right)\mathcal{N}_F(A_S, A_F) \\ \alpha_{2S} &= \mathcal{N}_S\left(A_S^{\text{ip}} + \frac{h}{2}\alpha_{1S}, A_F^{\text{ip}} + \frac{h}{2}\alpha_{1F}\right) \\ \alpha_{2F} &= \mathcal{N}_F\left(A_S^{\text{ip}} + \frac{h}{2}\alpha_{1S}, A_F^{\text{ip}} + \frac{h}{2}\alpha_{1F}\right) \\ \alpha_{3S} &= \mathcal{N}_S\left(A_S^{\text{ip}} + \frac{h}{2}\alpha_{2S}, A_F^{\text{ip}} + \frac{h}{2}\alpha_{2F}\right) \\ \alpha_{3F} &= \mathcal{N}_F\left(A_S^{\text{ip}} + \frac{h}{2}\alpha_{2S}, A_F^{\text{ip}} + \frac{h}{2}\alpha_{2F}\right) \\ \alpha_{4S} &= \mathcal{N}_S\left(\exp\left(\frac{h}{2}\mathcal{D}_S\right)\left(A_S^{\text{ip}} + h\alpha_{3S}\right), \exp\left(\frac{h}{2}\mathcal{D}_F\right)\left(A_F^{\text{ip}} + h\alpha_{3F}\right)\right) \\ \alpha_{4F} &= \mathcal{N}_F\left(\exp\left(\frac{h}{2}\mathcal{D}_S\right)\left(A_S^{\text{ip}} + h\alpha_{3S}\right), \exp\left(\frac{h}{2}\mathcal{D}_F\right)\left(A_F^{\text{ip}} + h\alpha_{3F}\right)\right) \\ A_S(z + h) &= \exp\left(\frac{h}{2}\mathcal{D}_S\right)\left(A_S^{\text{ip}} + \frac{h}{6}(\alpha_{1S} + 2\alpha_{2S} + 2\alpha_{3S})\right) + \frac{h}{6}\alpha_{4S} \\ A_F(z + h) &= \exp\left(\frac{h}{2}\mathcal{D}_F\right)\left(A_F^{\text{ip}} + \frac{h}{6}(\alpha_{1F} + 2\alpha_{2F} + 2\alpha_{3F})\right) + \frac{h}{6}\alpha_{4F}\end{aligned}\tag{8}$$

One should use the form of Equation 7 and finally apply $z' = z + \frac{h}{2}$ when deriving the above relations and remember to include the phase matching term when doing actual numerical calculation.

BIBLIOGRAPHY

- [1] S. Balac and F. Mahé, Embedded Runge–Kutta scheme for step-size control in the interaction picture method, *Computer Physics Communications* **184**, 1211 (2013).